<table>
<tr><th>Algorithm example</th><th>Equivalent in C language (with some C++ facilities)</th></tr>
<tr><td><pre>
Algo Exo2_3
Var
  x, Y : Integer
Begin
  Read (x)
  If x = -10 or x ≥ 0 and x ≠ 10 Then
    Y ← 1
  Else
    Y ← 2
  Endif
  Write(Y)
End
</pre></td><td><pre>
#include <iostream>
using namespace std;
int main()
{
  Int x, Y;
  cin >> x;
  if (x == -10 or x>=0 and x != 10)
  {
    Y = 1;
  }
  else
  {
    Y = 2;
  }
  cout << Y;
}
</pre></td></tr>
</table>

<table>
<tr><th>Algorithm example</th><th>Equivalent in C language (with some C++ facilities)</th></tr>
<tr><td><pre>
Algo Ex_2_13_c
Var
  A, B, C : Real
Begin
  Read (A, B)
  C ← A / B*B
  If C < 10 Then
    Write (A)
  Else
    If C ≤ 20 Then
      Write (C)
    Else
      Write (B)
    EndIf
  EndIf
End
</pre></td><td><pre>
#include <iostream>
using namespace std;
int main()
{
  float A, B, C;
  C = A / B * B;
  if(C <10)
  {
    cout << A;
  }
  else
  {
    if(C<=20)
    {
      cout << C;
    }
    else
    {
      cout << B;
    }
  }
}
</pre></td></tr>
</table>

Pseudo code example	Equivalent in C language (with some C++ facilities)
If Z mod 2 = 0 Then	**if(Z % 2 == 0)** **There are no semicolon after the if condition : If(Z % 2 == 0) ;**

<table>
<tr><th>Square root pseudo code example</th><th>Equivalent in C language (with some C++ facilities)</th></tr>
<tr><td><pre>
Algo Exmpl
  …
Begin
  …
  H ← sqrt(2 * F + 1)
  …
End
</pre></td><td><pre>
#include <iostream>
#include <math.h>
using namespace std;

int main()
{
  …
  H = sqrt(2*F+1);
  …
}
</pre></td></tr>
</table>

Do not consider possible user error cases in this series

Solution of Exercise 2.01:

Algo Exo2_1_a
Var
A, B, C : Integer
Begin
Read (A,B)
If A < 5 or B < 8 **Then**
C ← A + B * 2
Else
C ← A * 2 + B
EndIf
Write (C)
End
 	
For A=3 and B=6, C= 15
For A=3 and B=10, C= 23
For A=5 and B=10, C= 20
	Algo Exo2_1_b
	Var
	N, P : Integer
	Begin
	Read (N)
	If N mod 2 = 0 **Then**
	N ← N + 4
	EndIf
	P ← 1 + N * 3
	Write(P)
	End
	
	The mod operation gives the remainder of the division (13 mod 5 = 3)
	For N = 6, P = 31
	For N = 9, P = 28

Algo Exo2_1_c
Var
N, R : Integer
Begin
Read(N)
If N < 0 **Then**
R← 0
Else
If N < 10 **Then**
R← 1
Else
R← 2
EndIf
EndIf
Write(R)
End
 	
For N = 5, R = 1
For N = -13, R = 0
For N = 21, R = 2
	Algo Exo2_1_d
	Var
	A, B, C, D : Real
	Begin
	Read(A,B,C)
	D← A
	If B > D **Then**
	D← B
	EndIf
	If C > D **Then**
	D← C
	EndIf
	Write (D)
	End
	
	For A=2, B=15 and C=9, D = 15.0
	For A=0, B=-3 and C=12, D = 12.0
	For A=27,B=10 and C=19, D= 27.0
	
	- What does this algorithm do?

Solution of Exercise 2.08_c:

```
Algo Exo2_8_c
Var
  a, b, c, Sol1, Sol2, Delta : Real
Begin
    Read(a,b,c)
    Delta ← b*b – 4*a*c
    If Delta < 0 Then
      Write("No solution in R")
    Else
      If Delta = 0 Then
        Sol1 ← -b/(2*a)
        Write("Dual solution :", Sol1)
      Else
        Sol1 ← (-b+sqrt(delta))/(2*a)
        Sol2 ← (-b-sqrt(delta))/(2*a)
        Write("Two solutions :", Sol1, " " , Sol2)
      EndIf
    EndIf
End
```

For a=2, b=1 and c=-3, Two Solutions :1.0 -1.5

For a=1, b=-2 and c=1, Dual Sulution :1.0

For a=1,b=1 and c=1, No solution in R

Solution of Exercise 2.05:

```
Algo Exo2_5
Var
  X1, X2, X3, X4, X5, Max : Real
Begin
  Read(X1, X2, X3, X4, X5)
  Max← X1
  If X2>Max Then
    Max← X2
  EndIf
  If X3>Max Then
    Max← X3
  EndIf
  If X4>Max Then
    Max← X4
  EndIf
  If X5>Max Then
    Max← X5
  EndIf
  Write (Max)
End
```

Solution of Exercise 2.06:

```
Algo Exo2_6_a
Var
  Nb : Integer
Begin
  Read(Nb)
  If Nb mod 2 = 0 Then
    Write ("Even")
  Else
    Write ("Odd")
  EndIf
End
```

```
Algo Exo2_6_b
Var
  A, B, C : Integer
Begin
  Read(A,B,C)
  If A mod 5 = 0 Then
    Write (A)
  EndIf
  If B mod 5 = 0 Then
    Write (B)
  EndIf
  If C mod 5 = 0 Then
    Write (C)
  EndIf
End
```

Exercise 2.10 (Home Work):

The calendar year is divided into 12 months numbered 1 to 12 with the following principles:
- months 1, 3, 5, 7, 8, 10 and 12 consist of 31 days.
- months 4, 6, 9 and 11 consist of 30 days.
- month 2 consists of 29 days if the year is a **leap year** and 28 days otherwise.

Note: a year (AD : After Christ) is a leap year if :
  - it is divisible by 4 and not divisible by 100, or
  - it is divisible by 400.

- Write a program which has a month and a year as input and which displays the number of days in this month.